

THE POLICY ENDOGENEITY TRAP & ACTIVE DATA FLYWHEEL

Preventing Distribution Collapse by Decoupling Autonomous Recovery from Intervention Demonstrations

× THE NAÏVE INTERVENTION TRAP

Retraining directly on raw human takeovers

1. Policy drifts near ODD boundary

Accumulating compounding error $x_t \sim d_\pi$



2. Human takes over aggressively

Discontinuous recovery action u_{human}



3. Policy trained on takeover data

Model learns that 'danger implies human assist'



4. Autonomous recovery collapses

Agent freezes or drives deeper into hazard

✓ THE DEFENSIVE DATA FLYWHEEL

DAgger with Truncated Demonstrations & Counterfactuals

1. State Visitation Distribution d_π

Agent executes in world under policy π_θ



2. Expert Labeling on Policy States

Supervisor labels corrective action $u^*(s)$ for $s \sim d_\pi$



3. Policy Lineage & Hash Auditing

Dataset indexed by commit hash & seed



4. Monotonic ODD Expansion

Empirical safety margin improves across shadow fleet